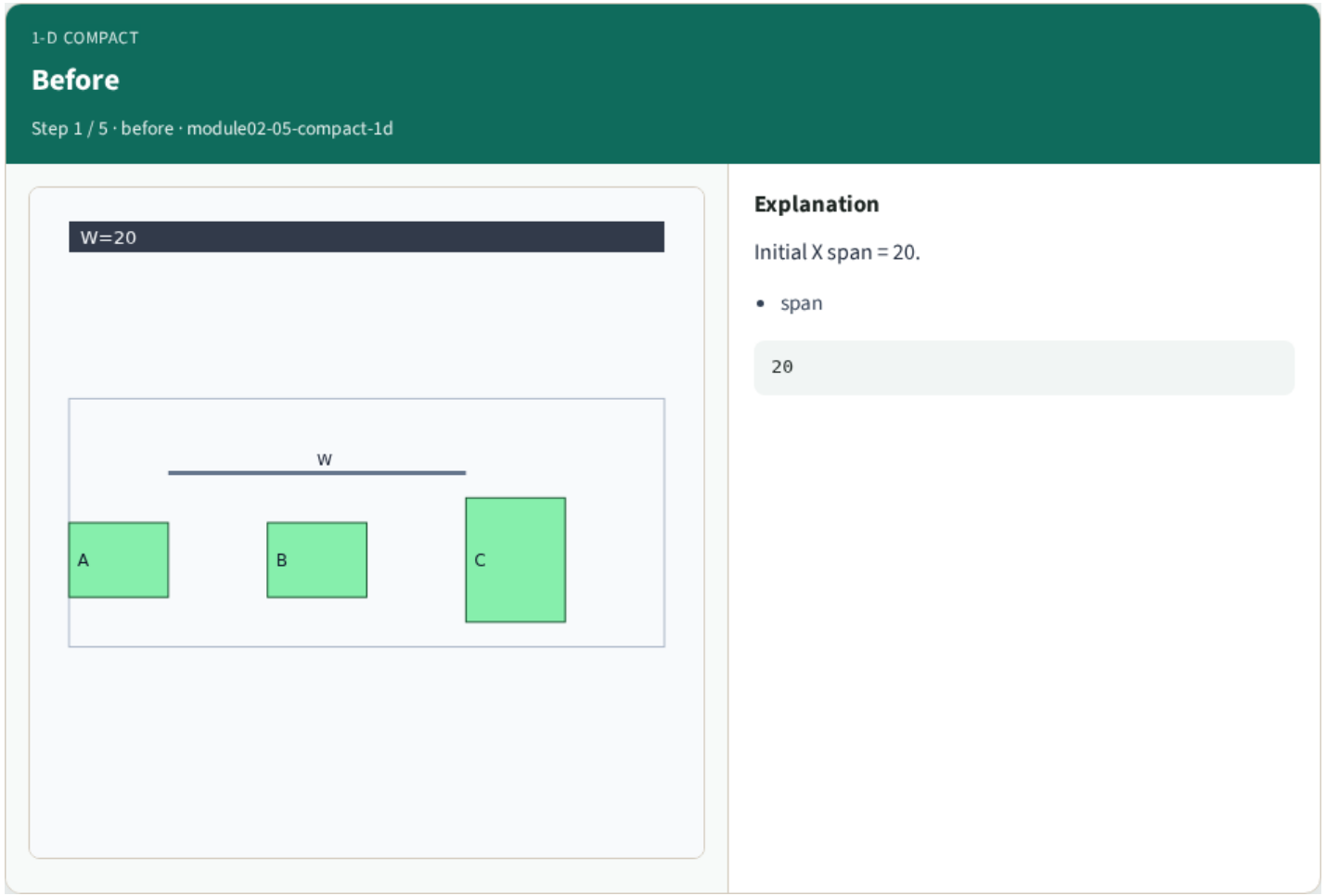
# 1-D compact

One axis at a time: build graph, longest-path, rewrite coordinates

# The idea

- GOLDEN: X compaction before_span=20, after_span=14, improved=True

# Before



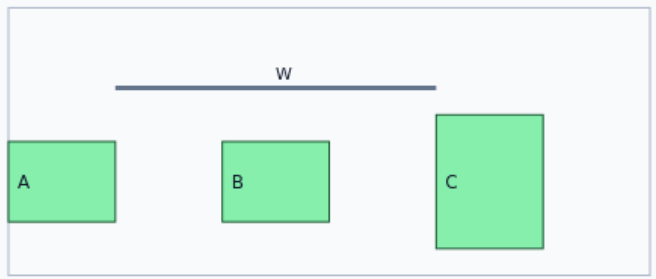

Before

# Run

1-D COMPACT

## Run

Step 2 / 5 · run · module02-05-compact-1d




**Explanation**

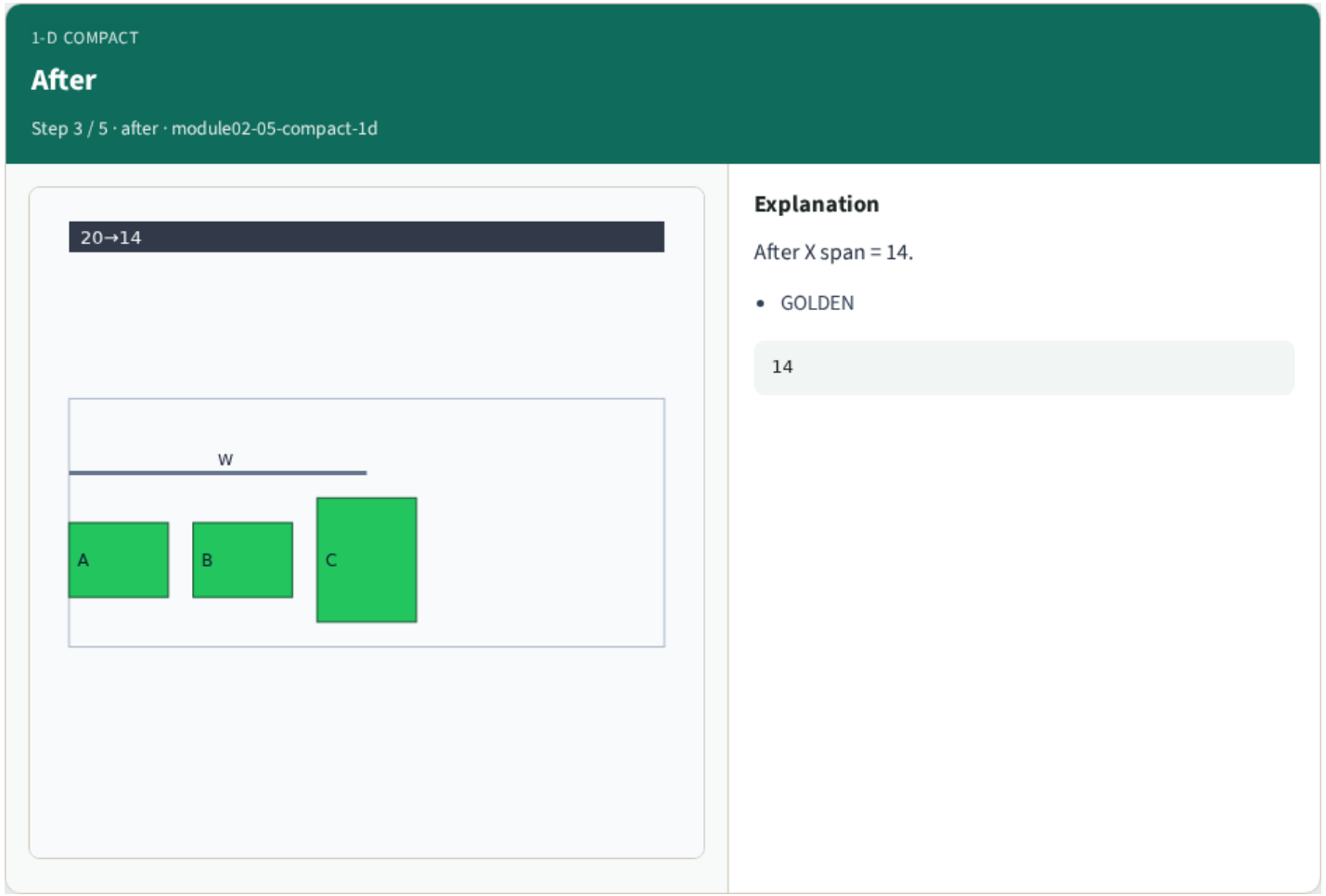
compact_1d(layout, 'x').

- API

```
X
```

Run

# After


1-D COMPACT
After
Step 3 / 5 · after · module02-05-compact-1d
20→14
W
A
B
C
Explanation
After X span = 14.
GOLDEN
14

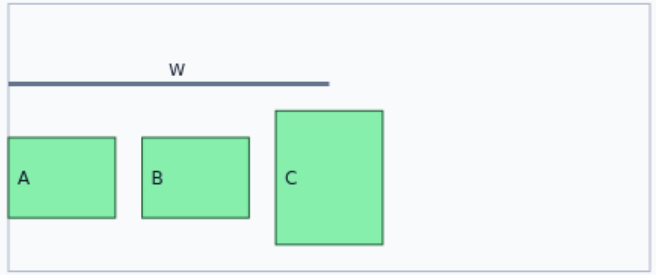
After

# Gaps

1-D COMPACT

## Gaps

Step 4 / 5 · gaps · module02-05-compact-1d




**Explanation**

Whitespace between A–B–C shrinks to min_gap.

- pack

```
gap=1
```

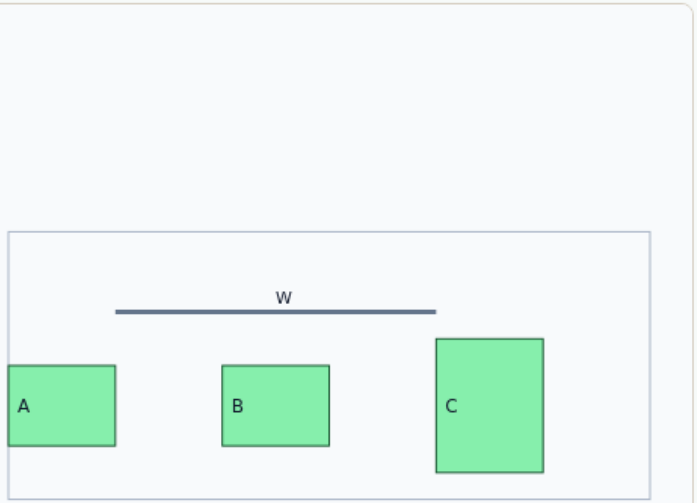
Gaps

# Y untouched

1-D COMPACT

**Y untouched**

Step 5 / 5 · y · module02-05-compact-1d




**Explanation**

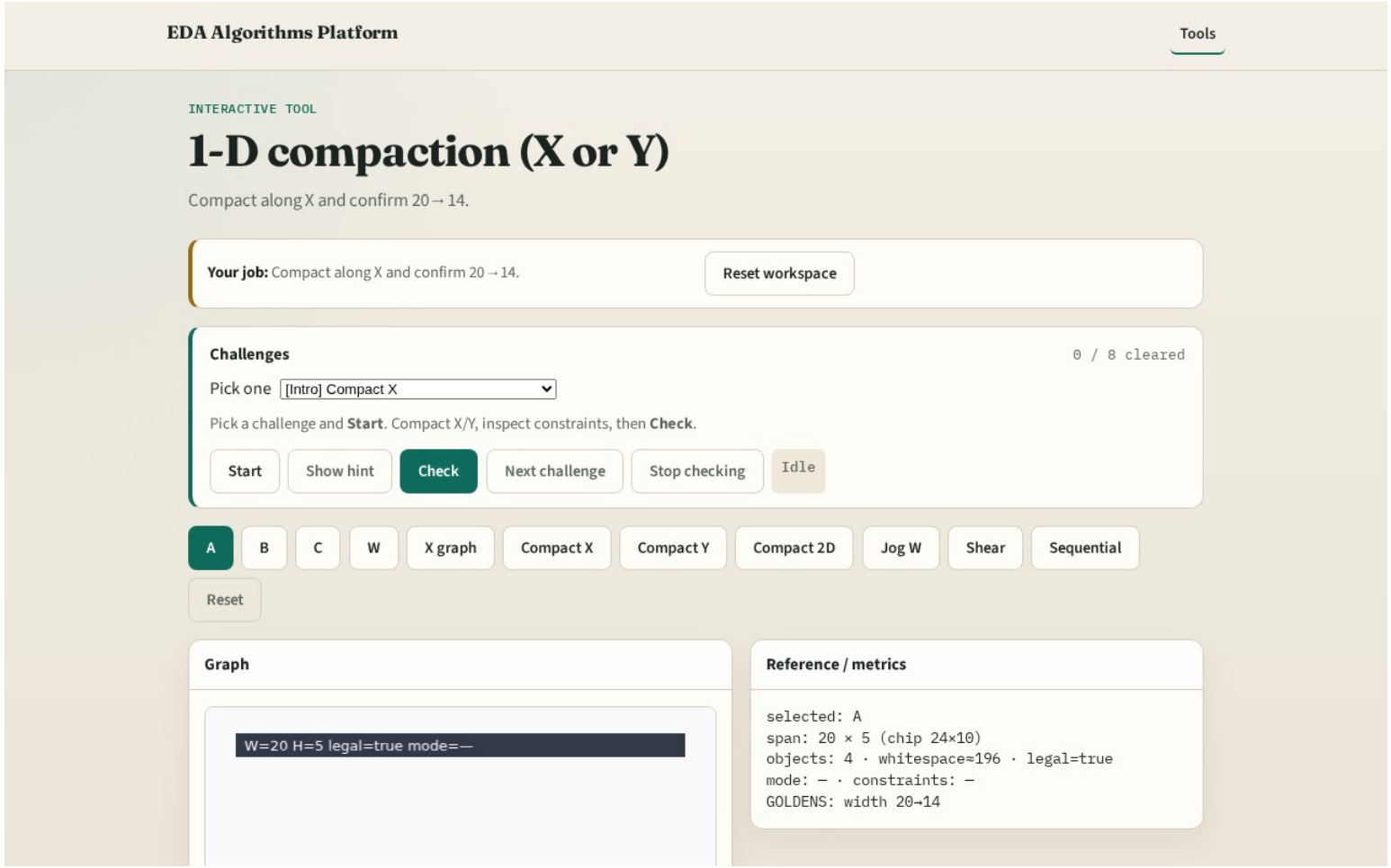
Y coordinates unchanged in a pure X pass.

- axis

```
X only
```

Y untouched

# Browser lab track




Browser lab starter

# Implement track

- Call compact_1d(DATA,'x'); assert after_span==14

# Pitfalls

- Expecting Y to change
- Ignoring improved flag

# Your turn

- Next: 2-D compaction